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Variably irradiated tidally locked planets

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ABSTRACT

Temporally ~~modulated~~ stellar heating affects the climate of pseudo-synchronously rotating planets on eccentric orbits and those orbiting gravity-darkened stars. We study the atmospheric response of variably irradiated tidally locked exoplanets using a shallow water model forced toward a time-dependent radiative equilibrium. Linearizing the equations, we derive analytic expressions for the amplitude and phase lag of the atmospheric response in terms of dynamical parameters, including wave, radiative, and drag timescales, effectively reducing seven parameters to two. In the limit of steady forcing, our results recover known features of synchronously rotating planets, ~~validating the model~~. We find close matches of the analytical results with linear and nonlinear simulations albeit with minor quantitative differences. At ~~low~~ drag/dissipation, when the wave timescale matches the radiative timescale, the hotspot location shows periodic oscillations corresponding to the variation of instellation moving between the substellar and antistellar points~~, having~~ imprints on the phase curve of such planets.

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1. INTRODUCTION

Many close-in exoplanets are expected to be *tidally locked* to their stars, so that one hemisphere receives continuous stellar irradiation while the opposite hemisphere remains in perpetual darkness. In such cases a permanent dayside and nightside develop, leading to extreme longitudinal heating contrasts (R. Pierrehumbert & M. Hammond 2019). This intense day–night thermal forcing drives a ~~characteristic circulation: broad~~ equatorial superrotation and planetary-scale standing waves ~~arise to~~ transport heat. Analytical and numerical models show that standing equatorial waves forced by the day–night heating can pump eastward momentum ~~into the equator,~~ naturally producing a wide superrotating jet and an eastward-shifted thermal hotspot (A. P. Showman & L. M. Polvani 2011; D. Perez-Becker & A. P. Showman 2013). For example, Perez-Becker & Showman (2013) used a ~~two-~~layer shallow-water model to demonstrate that strong stellar forcing (short radiative time) leads to nearly direct day–night flow with order-unity temperature contrasts, whereas weaker forcing yields banded zonal flows with smaller contrasts (D. Perez-Becker & A. P. Showman 2013). These results are borne out by 3D circulation models and reviewed in detail (R. Pierrehumbert & M. Hammond 2019).

However, this picture assumes constant stellar forcing. In many systems the irradiation varies periodically. Examples include:

- **Eccentric orbits / pseudo-synchronous spin:** HD 80606b ($e \approx 0.93$) receives short, intense bursts of heating near each periastron (J. Langton & G. Laughlin 2008).
- **Gravity-darkened stars:** KELT-9 is a very rapidly rotating star whose poles are much hotter than its equator (von Zeipel effect), so KELT-9b sees a latitude-dependent flux pattern that rotates with the star (K. D. Jones et al. 2022).
- **Pulsating/variable stars:** WASP-33b orbits a δ-Scuti pulsator, causing its incident flux to oscillate on timescales comparable to the orbital period (C. von Essen et al. 2020).

In each of these cases the pattern of heating on the planet changes over time, even if the planet's spin re-

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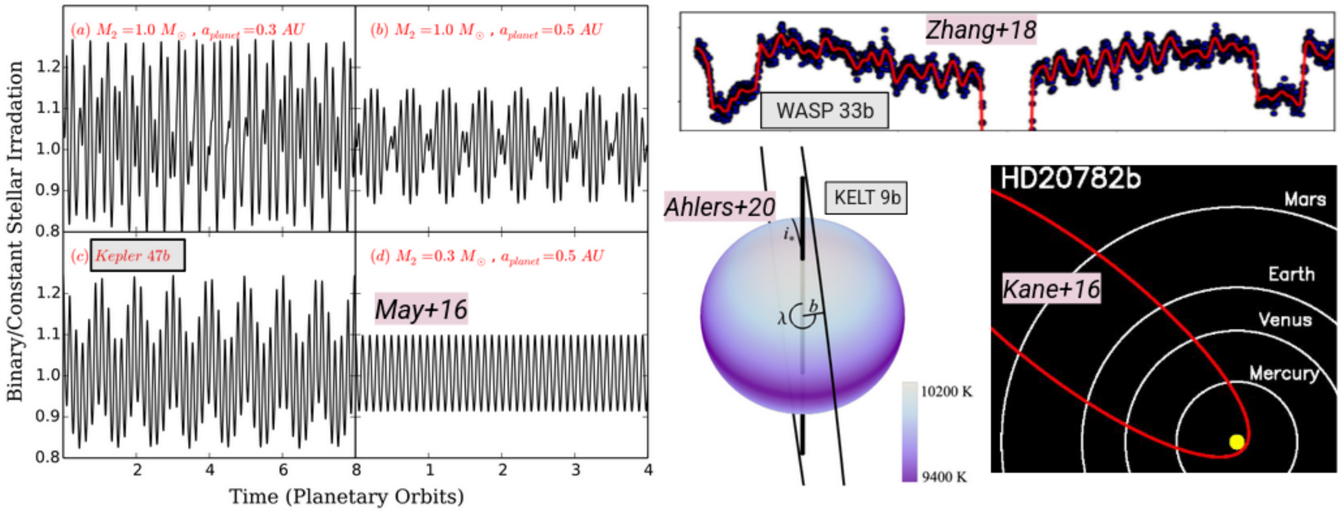

Figure 1. Different kinds of variably irradiated systems, including circumbinary planets (Kepler-47b), tidally-locked planets around variable stars like the Delta Scutti variables (WASP-33b), planets in polar orbits around gravity darkened stars (KELT-9b), and eccentric/pseudo-synchronous planets (HD20782b).

mains synchronized on average. This time-dependent forcing can significantly modify the atmospheric winds and temperature distribution compared to the steady case, and thus the observed phase curve.

Several studies have begun to explore time-varying irradiation. GCM simulations have found that eccentric hot Jupiters still develop strong equatorial jets, but the timing of peak infrared emission can shift relative to periastron (showing "ringing" in phase curves) depending on orbital geometry (T. Kataria et al. 2013). Compressible shallow-water simulations showed that HD 80606b's atmosphere is dominated by the periastron heating pulse, driving high-speed turbulent flows and long-lived vortices (J. Langton & G. Laughlin 2008). Recently, novel tidally locked circulation regimes have been demonstrated on asynchronous planets when the instellation variation period nearly matches the planet's day fixing the hotspot to a fixed longitude (D. Banik 2025). However, a generalized analytic theory tested against numerical simulations for the response of a planet to oscillatory irradiation is still lacking.

We use the 1.5 layer standard shallow water setup study variably irradiated tidally-locked planets, although the system could in principle be extended to other cases. Through linearized analysis we find analytical expressions for the steady-state response, limits on linear and nonlinear behaviour, and the time-varying response as a function of known system parameters. We ~~theoretically~~ predict that tidally-locked planets ~~are~~ damped harmonic oscillators and verify the claim with linear and nonlinear simulations in Dedalus3 (K. J. Burns et al. 2020), finding close matches for the overall planetary day-night contrast. In the limit of zero forcing frequency our solution reduces to the standard tidally locked solution of D. Perez-Becker & A. P. Showman (2013). However, the dynamics captured in simulations show interesting forms of variability triggered by changing instellation, specifically related to hotspot oscillations. Previous magnetohydrodynamic studies have shown strong toroidal magnetic fields ~~to~~ drive periodic east–west hotspot reversals (T. M. Rogers & A. P. Showman 2014; T. M. Rogers & T. D. Komacek 2014; A. W. Hindle et al. 2019, 2021a,b). We note that the specific hotspot oscillations reported in this work is not entirely new; the closest analog to this phenomena has been reported in GCMs of hot Jupiters on eccentric orbits that show the hotspot location varying periodically (T. Kataria et al. 2013; N. K. Lewis et al. 2014). However, the conditions under which such oscillations grow and stabilize are a direct output our generalized theory and hence novel. We isolate global-scale hotspot oscillations that arise purely from the wave-dynamical response of an hydrodynamic atmosphere to a periodically time-varying instellation, without invoking magnetic or instrinsic variability effects. While our forcing is simply sinusoidal similar to D. Banik (2025), the generalized theory could in principle be extended to any form of instellation variation, including eccentric orbits, gravity-darkened stars, and variable stars. Figure 1 shows a few examples of such systems.

The remainder of this paper is organized as follows. In Section 2 we develop the linearized shallow water framework for variably irradiated tidally locked planets. Starting from the full nonlinear equations, we derive scaling relations for the steady-state height contrast and characteristic wind speed, identify the conditions under which the linear approximation holds, and solve the time-dependent equations analytically for sinusoidal instellation forcing. This yields closed-form expressions for the amplitude and phase lag of the atmospheric response as functions of two nondimensional parameters, K and W, with a third parameter X entering at second order when inertial effects are retained. In Section 3 we describe the numerical simulations used to validate the theory, covering Earth-like and hot Jupiter parameter regimes across 200 ~~runs~~ spanning both linear and nonlinear forcing amplitudes. In Section 4 we discuss the direction of hotspot oscillations as a function of rotation rate, examine the effect of nonlinearities on suppressing westward oscillations, and construct synthetic phase curves illustrating the observable consequences of variable irradiation. We summarize our main conclusions in Section 5.

2. LINEARIZED SHALLOW WATER THEORY

We utilize the same spherical (λ, ϕ) two-layer shallow water model as in D. Banik (2025) to simulate the atmosphere of variably irradiated tidally-locked planets. ~~For details the reader is refered to that paper.~~ The shallow water model offers the benefit of efficient parameter space survey in addition to capturing leading order circulation physics, and ~~has~~ been extensively used in ~~literature before~~ (D. Perez-Becker & A. P. Showman 2013; K. Ohno & X. Zhang 2019; J. Penn & G. K. Vallis 2017). The model features an active upper layer $h(\lambda, \phi, t)$ over an inert layer of infinite depth. The governing equations for momentum and mass balance are,

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} + f\mathbf{k} \times \mathbf{u} + g\nabla h = \mathbf{R} - \frac{\mathbf{u}}{\tau_{\rm drag}} \quad (1)$$

$$\frac{\partial h}{\partial t} + \nabla \cdot (h\mathbf{u}) = \frac{h_{\rm eq} - h}{\tau_{\rm rad}} \equiv Q \quad (2)$$

$$\mathbf{R} = -\frac{\mathbf{u}}{h} \cdot \frac{Q + |Q|}{2} \quad (3)$$

where g is the acceleration due to gravity on the planet, $\mathbf{u}$ is horizontal velocity, $f = 2\Omega \sin\phi$ is the Coriolis parameter, $\tau_{\rm drag}$ is the Rayleigh drag timescale, and $\tau_{\rm rad}$ is the Newtonian cooling timescale. The term $\mathbf{R}$ ensures conserved mass/momentum exchanges between the two layers, emergence of equatorial superrotation, and ensures independence from initial conditions (A. P. Showman & L. M. Polvani 2010a; A. P. Showman & L. Polvani 2011; B. Liu & A. P. Showman 2013). Fluid height acts as a proxy for temperature.

2.1. *Steady-state Scaling Analysis*

We carry out scaling analysi~~s~~ similar to D. Banik et al. (2025). For that, we linearize the system with the following assumptions:

- The basic state ~~are~~ zero background flow ~~for~~ velocity $\mathbf{u}_0 = 0$ and the nightside height H ~~for height.~~

- Small perturbations are added on top of the base state i.e. $\mathbf{u} = \cancel{0} + \mathbf{u}', \quad h = H + h'$, with $|\mathbf{u}'| \ll \sqrt{gH} = c$, or the gravity wave speed of the shallow atmosphere, and $|h'| \ll H$.

- We decompose $h_{\rm eq} = H + h'_{\rm eq}$, not for the purpose of linearizing but to express the equation in terms of the variable $h'_{\rm eq}$, the deviation from background height.

- We also neglect the nonlinear vertical exchange term $\mathbf{R}$ to simplify the system.

- Finally, we combine the Coriolis and drag terms to form ~~the~~ effective drag approximation ($\mathbf{k} \times \mathbf{u}' \approx \sin\phi\, \mathbf{u}', \quad \sin\phi \approx \sin\phi_0, \quad \tau_{\rm eff}^{-1} = f \sin\phi_0 + \tau_{\rm drag}^{-1}$). Thus, the combined term is $\tau_{\rm eff}^{-1}\, \mathbf{u}'$. We acknowledge that rotational and drag effects can be dramatically different especially considering latitudinal variations. However, at first order both terms contribute similarly to heat transport (D. Perez-Becker & A. P. Showman 2013; D. Banik 2025).

We first derive the scales for the unknown quantities height and velocity from the linearized treatment of the steady-state by dropping the time derivatives. Then we use these to derive conditions for nonlinearity through order-of-magnitude comparisons. We define the following characteristic scales: horizontal length $L \sim a$ (planetary radius), velocity U, height perturbation Δh, forcing $\Delta h_{\rm eq}$, and nightside thickness H. The linearized equations of motion produce the following momentum and mass balances,

$$0 = -g\nabla h' - \frac{\mathbf{u}'}{\tau_{\rm eff}} \implies 0 \sim -g\frac{\Delta h}{a} - \frac{U}{\tau_{\rm eff}} \quad (4)$$

$$0 = -H\nabla \cdot \mathbf{u}' + \frac{h'_{\rm eq} - h'}{\tau_{\rm rad}} \implies 0 \sim -H\frac{U}{a} + \frac{\Delta h_{\rm eq} - \Delta h}{\tau_{\rm rad}} \quad (5)$$

From the momentum balance (4), we find the velocity scale as,

$$U \sim \frac{g\Delta h \tau_{\rm eff}}{a} \quad (6)$$

We use the gravity wave speed $c = \sqrt{gH}$ to define the wave timescale as $\tau_{\rm wave} = a/c$. Substituting the above into the continuity balance (5) we get

$$\frac{\Delta h}{\Delta h_{\rm eq}} \sim \left(1 + \frac{\tau_{\rm rad}\tau_{\rm eff}}{\tau^2_{\rm wave}}\right)^{-1} \text{ or } \Delta h = \frac{\Delta h_{\rm eq}}{1+1/K} \qquad (7)$$

where $K = \tau^2_{\rm wave}/(\tau_{\rm rad}\tau_{\rm eff})$. Thus, we see that the global contrast Δh is dependent only on two parameters, the forcing $\Delta h_{\rm eq}$ and K, the dimensionless ratio of wave and damping (radiative and momentum) timescales. This is commensurate with the scaling law for hot Jupiters in D. Perez-Becker & A. P. Showman (2013), except it is a more condensed version combining the effects of rotation and drag in a simplified form. The characteristic velocity U is of the form,

$$U \sim \frac{a\tau_{\rm eff}}{H\tau^2_{\rm wave}} \cdot \frac{\Delta h_{\rm eq}}{1+1/K} \qquad (8)$$

This is the velocity scaling corresponding purely to thermal forcing, and has a similar structure to that of momentum forcing derived in T. D. Komacek (2025).

2.1.1. *Conditions for linearity*

The linear approximation holds only when the nonlinear terms in the momentum and continuity equations are negligible compared to the corresponding linear terms.

- Height Linearity Condition: The thickness advection $\nabla\cdot(h\mathbf{u}')$ which scales as $\Delta hU/a$ must be small compared to the mean mass flux $H\nabla\cdot\mathbf{u}' \sim HU/a$. This is satisfied when
$$\frac{\Delta h}{H} = \frac{\Delta h_{\rm eq}/H}{1+1/K} \ll 1. \qquad (9)$$

- Momentum Linearity Condition: From the above scalings, the advection term $(\mathbf{u}'\cdot\nabla)\mathbf{u}'$ scales as U^2/a and must be small compared to that of drag $U/\tau_{\rm eff}$. Substituting U from (8) we get,
$$\frac{U^2/a}{U/\tau_{\rm eff}} = \frac{\tau^2_{\rm eff}}{\tau^2_{\rm wave}} \cdot \frac{\Delta h_{\rm eq}/H}{1+1/K} \ll 1. \qquad (10)$$
In a different, but familiar geophysical fluid dynamics context, this corresponds to the "frictional" Rossby number (rotation plus drag) being smaller than 1, such that inertia/ material acceleration terms cannot dominate the dynamics.

D. Perez-Becker & A. P. Showman (2013), henceforth also refered to as PBS13, set the ratio of forcing to the nightside height ($\Delta h_{\rm eq}/H$) to 0.001 and 1, to cover linear and nonlinear regimes ~~of operation.~~ However, that is not sufficient to ensure either linear or nonlinear behavior. Both (10) and (9) must be satisfied, out of which the momentum condition, unlike the mass condition, depends on $\tau^2_{\rm eff}/\tau^2_{\rm wave}$ in addition to the forcing $\Delta h_{\rm eq}$ and K. These conditions have not been previously derived for a forced shallow water model of a tidally-locked planetary atmosphere. figure 2 plots the LHS of the linearity conditions as a function of K for two different values of $\Delta h_{\rm eq}/H$, and two cases of planetary parameters, one corresponding the typical hot Jupiter assumed in D. Perez-Becker & A. P. Showman (2013) for which $X = 3$, and the other assumed arbitrarily for this work. Here, $X = \tau_{\rm wave}/\tau_{\rm rad}$. Comparing figures 3 and 4 in PBS13, we find that the difference between linear and nonlinear flows is more pronounced at low drag, which here corresponds to low K (high $\tau_{\rm drag}$) values, making their results agree with our scalings.

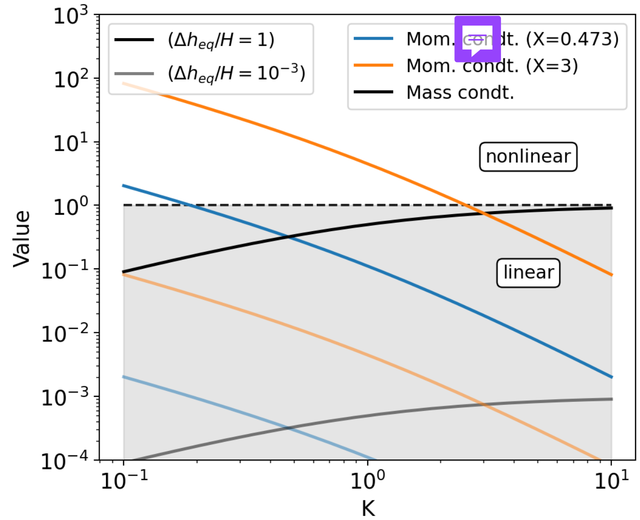

Figure 2. Magnitude of linearity constants in inequalities (10) and, (9) with K for different system parameters assumed in this work. Note that $X = \tau_{\rm wave}/\tau_{\rm rad}$.

Figure 3 shows how the nondimensional forcing amplitude $\Delta h_{\rm eq}/H$ varies as a function of the parameter K for a suite of numerical experiments spanning Earth-like and hot-Jupiter-like parameter choices performed later in the paper. Data points correspond to simulation results (circles for the linear regime, squares for nonlinear runs) while the solid lines show the theoretical prediction from the linearized shallow-water balance. The plot demonstrates good agreement between theory and simulations in the linear regime, and systematic departures in the nonlinear cases especially at low K (weak drag or rotation), highlighting the role of nonlinearity in enhancing the height contrast. (What does PBS13 other literature say?)

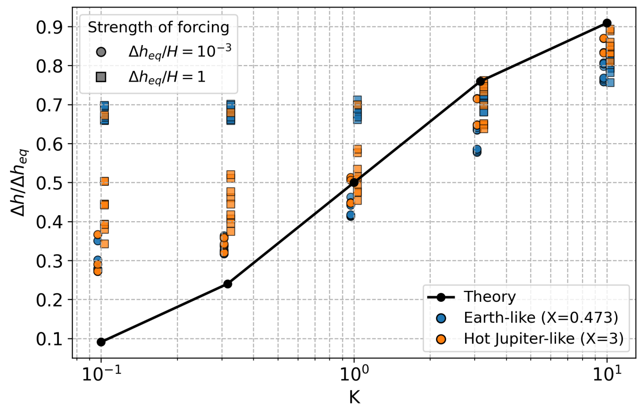

Figure 3. Variation of $\Delta h_{\rm eq}/H$ with K for different Earth--like and Hot Jupiter planetary system parameters from all simulations performed in the following section compared with theoretical predictions. The linear (circles) and non-linear (squares) regime simulations are horizontally offset by small quantity for the ease of visualization. Linear regime simulations match the theory very well, while the nonlinear regime simulations show quantitative differences, as expected.

2.2. *Linearized first-order dynamics*

When both the above conditions are satisfied, we may drop the corresponding nonlinear terms. In this section we consider the linearized limit in an attempt to find analytical solutions to relevant quantities like ~~temperature~~ contrast and average wind speed. Since we are investigating the effect of changing irradiation on the atmospheric height we must retain the unsteady term in equation (2). We begin with the assumption of a balanced flow originating from equation (1) where the pressure gradient is balanced by effective drag term. Ignoring nonlinear advection and unsteady terms gives us an ~~equation~~ similar to (4).

$$\frac{1}{\tau_{\rm eff}}\,\mathbf{u}' + g\,\nabla h' = \mathbf{0} \quad\Longrightarrow\quad \mathbf{u}' = -\,g\,\tau_{\rm eff}\,\nabla h'. \tag{11}$$

Keeping $\partial_t h'$ in the mass balance and linearizing yields

$$\frac{\partial h'}{\partial t} + H\,\nabla\cdot\mathbf{u}' = \frac{h'_{\rm eq} - h'}{\tau_{\rm rad}}. \tag{12}$$

From this linearized height equation, we may compare the scales of the unsteady and the height advection (mass flux) terms to find the following timescale for the flow in general,

$$\frac{\Delta h}{\tau_{\rm o}} \sim \frac{HU}{a} \implies \tau_{\rm o} \sim \tau_{\rm adv}\cdot\frac{\Delta h}{H} = \frac{\tau_{\rm wave}^2}{\tau_{\rm eff}} \tag{13}$$

where the advection timescale ($\tau_{\rm adv} = a/U$) has been modified by the linearity constant in (9).

Taking ~~divergence on~~ both sides of equation (11) we get $\nabla\cdot\mathbf{u}' = -\,g\,\tau_{\rm eff}\,\nabla^2 h'$ which can then be substituted in equation (12) to obtain the following PDE for height,

$$\frac{\partial h'}{\partial t} - gH\,\tau_{\rm eff}\,\nabla^2 h' + \frac{h'}{\tau_{\rm rad}} = \frac{h'_{\rm eq}(\lambda,\phi,t)}{\tau_{\rm rad}}. \tag{14}$$

This is a generalized first order equation in time representing the dynamics of planetary response to any form of spatially and temporally varying radiation. Next we shall investigate the behavior of the system.

Let us assume invariance in the meridional direction (ϕ) and reduce the spatial dimension of the problem to one. For simplicity, we shall also move to arc-length coordinates and replace λ by x. Additionally, we assume the forcing profile to be a simple cosine in x as follows $h'_{\rm eq} = \Delta h_{\rm eq}\,f(t)\cos(kx)$ (Add figure showing lat, lon, cosine, projection etc.), where $f(t)$ is some function of time.

We seek solutions of the form $h'(x) = \Delta h(t)\cos(kx)$, where $\Delta h(t)$ is the time-varying amplitude to be determined. This solution exists under the assumption of pure substellar to antistellar point flow. This results in the following linear damped forced ODE for the planetary ~~heat/~~height contrast,

$$\frac{d}{dt}\Delta h + \left(\frac{1}{\tau_{\rm rad}} + gHk^2\,\tau_{\rm eff}\right)\Delta h = \frac{\Delta h_{\rm eq}\,f(t)}{\tau_{\rm rad}} \tag{15}$$

Having solved for Δh, we may substitute h' ~~in~~ equation (11) to uniquely determine the magnitude of the velocity field. Note that the velocity field varies as a sine from the substellar point due to the gradient operator on h', implying that the highest and lowest temperature (max/min h') points or the hotspot and the coldspot are regions of stagnation and the winds achieves maximum value near the day-night divide i.e. the terminator.

If we take $f(t)$ to be unity and look for steady-state solutions, we recover the relation between planetary response and forcing as obtained from scaling relations in equation (8) ~~which is~~ as expected.

2.2.1. *Variable instellation*

In this section, we shall focus on systems with variable instellation. Even though the real nature of variability can be significantly complicated as shown in figure 1, we start with simpler forms. We assume the forcing to be a simple sine superimposed on a background value and set $f(t) = (1+F\sin\omega t)$, where ω is the instellation variation frequency. Beyond the transient, we obtain solutions of the form, $\Delta h(t) = \Delta h_0 + \Delta h_{\rm amp}\cos(\omega t - \phi)$ to get,

$$\frac{\Delta h_0}{\Delta h_{\rm eq}} = \left(1 + \frac{\tau_{\rm rad}\tau_{\rm eff}}{\tau_{\rm wave}^2}\right)^{-1} = \frac{1}{1+1/K}, \tag{16}$$

$$\frac{\Delta h_{\mathrm{amp}}}{\Delta h_{\mathrm{eq}}} = F\left[\left(1+\frac{\tau_{\mathrm{rad}}\tau_{\mathrm{eff}}}{\tau_{\mathrm{wave}}^2}\right)^2 + (\omega\tau_{\mathrm{rad}})^2\right]^{-1/2}$$
$$= F\left[(1+1/K)^2 + W^2\right]^{-1/2}, \text{and} \qquad (17)$$

$$\phi = \tan^{-1}\left(\frac{\omega\tau_{\mathrm{rad}}}{1+\dfrac{\tau_{\mathrm{rad}}\tau_{\mathrm{eff}}}{\tau_{\mathrm{wave}}^2}}\right) = \tan^{-1}\left(\frac{W}{1+1/K}\right). \qquad (18)$$

where $W = \omega\tau_{\mathrm{rad}}$ is the nondimensional instellation variation frequency. We also define ~~some~~ other nondimensional parameters $X = \tau_{\mathrm{wave}}/\tau_{\mathrm{rad}}$, $Y = \tau_{\mathrm{wave}}/\tau_{\mathrm{drag}}$, and $Z = a/L_d$ representing radiative, drag and rotational strengths as relevant to the global flow. Note here that $K = X(Y+Z)$. We see that the final expressions cruicially depend on two parameters K and W only. Thus, the dimensionality of the parameter space is reduced from seven dimensional parameters ($f, g, a, H, \tau_{\mathrm{rad}}, \tau_{\mathrm{drag}}$, and ω) to two just nondimensional parameters. Here, K represents the planet's inherent capacity to prevent heat redistribution, while W is the nondimensional frequency of instellation variation.

The first three panels in figure 4 show the time-mean height ~~magnitude~~, its amplitude and phase of the planetary response to instellation variation in the $K-W$ space. The constant part of the planetary contrast is the same as obtained before and is unaffected by a change in ω. A large value of K corresponds to a planet that is strongly dragged, fast rotating, strongly radiated or any combination of them, all of which helps prevent heat redistribution. On the contrary, a planet with fast wave ~~physics~~ has a low K, and has better heat redistribution leading to an erasure of contrasts. The constant part of the contrast normalized by the forcing amplitude is thus seen to increase with K peaking at 1.

The amplitude of variation in planetary temperature behaves in a similar manner, except it is also affected by the frequency of instellation variation. When W or ω is large, the instellation fluctuates fast and the planet does not get enough time to adjust to the radiation leading to a muted amplitude irrespective of the value of K, while low W allows the system to respond appropriately, leading to higher amplitudes.

Finally, the time phase lag of the planetary response is the highest ($=\pi/2$, the theoretical maximum) for strongly dragged systems under rapid instellation variation. Note that this phase is not the hotspot offset from the substellar point but the time difference between the peak of instellation and planetary maxima. More discussion on the hotspot offset and associated variability is included in a later section.

2.3. *Linearized second-order dynamics*

While the above first-order treatment with balanced flows is the most minimal model to capture time-variability, we recognize that the unsteady term in the momentum equation (1) need not be negligible. In fact, the condition for the unsteady term to be significant may be obtained by setting the scale for the term to be much greater than the effective drag term as below,

$$\frac{U}{\tau_{\mathrm{o}}} \gg \frac{U}{\tau_{\mathrm{eff}}} \quad \text{or} \quad \tau_{\mathrm{o}} \ll \tau_{\mathrm{eff}}. \qquad (19)$$

Substituting the timescale τ_{o} from equation (13), we get the condition to include the unsteady term as,

$$\tau_{\mathrm{wave}} \ll \tau_{\mathrm{eff}}. \qquad (20)$$

This implies that that the drag or rotation of the system must be weak enough such that waves can dominate. In the opposite limit, gravity wave perturbations generated by heating are quickly damped preventing their effect from reverberating across the planet. When waves dominate new behavior emerges through constructive or destructive interference leading to resonances, as we shall see further. Assuming a cosine spatial structure along latitudes as before, we now simplify the mass and momentum balances keeping both unsteady terms to arrive at a second order ODE for the height as follows,

$$\Delta\ddot{h} + \left(\frac{1}{\tau_{\mathrm{rad}}}+\frac{1}{\tau_{\mathrm{eff}}}\right)\Delta\dot{h} + \left(\frac{1}{\tau_{\mathrm{wave}}^2}+\frac{1}{\tau_{\mathrm{rad}}\tau_{\mathrm{eff}}}\right)\Delta h$$
$$= \Delta h_{\mathrm{eq}}\left[\frac{1}{\tau_{\mathrm{rad}}}\dot{f(t)} + \frac{1}{\tau_{\mathrm{rad}}\tau_{\mathrm{eff}}}f(t)\right]. \qquad (21)$$

We compare this equation with $\Delta\ddot{h} + \gamma\Delta\dot{h} + \Omega_{\mathrm{o}}^2\Delta h = F(t)$, which is the equation for a forced damped harmonic oscillator as also identified in O. Shamir et al. (2023). This equation admits wave like solutions in time even without a sinusoidal forcing and has a natural frequency $\Omega_{\mathrm{o}} = \sqrt{\tau_{\mathrm{wave}}^{-2} + \tau_{\mathrm{rad}}^{-1}\tau_{\mathrm{eff}}^{-1}}$, and damping $\gamma = \tau_{\mathrm{rad}}^{-1} + \tau_{\mathrm{eff}}^{-1}$. The constant part of the solution remains the same as before, but the amplitude and phase are modified as follows,

$$\frac{1}{F}\cdot\frac{\Delta h_{\mathrm{amp}}}{\Delta h_{\mathrm{eq}}} = \frac{\dfrac{1}{\tau_{\mathrm{rad}}}\sqrt{\dfrac{1}{\tau_{\mathrm{eff}}^2}+\omega^2}}{\sqrt{(\Omega_{\mathrm{o}}^2-\omega^2)^2+(\gamma\omega)^2}}$$
$$= \frac{\sqrt{K^2+W^2X^4}}{\sqrt{(1+K-W^2X^2)^2+W^2(X^2+K)^2}} \qquad (22)$$

$$\tan\phi = \frac{\gamma\omega}{\Omega_{\mathrm{o}}^2-\omega^2} = \frac{W(X^2+K)}{1+K-W^2X^2} \qquad (23)$$

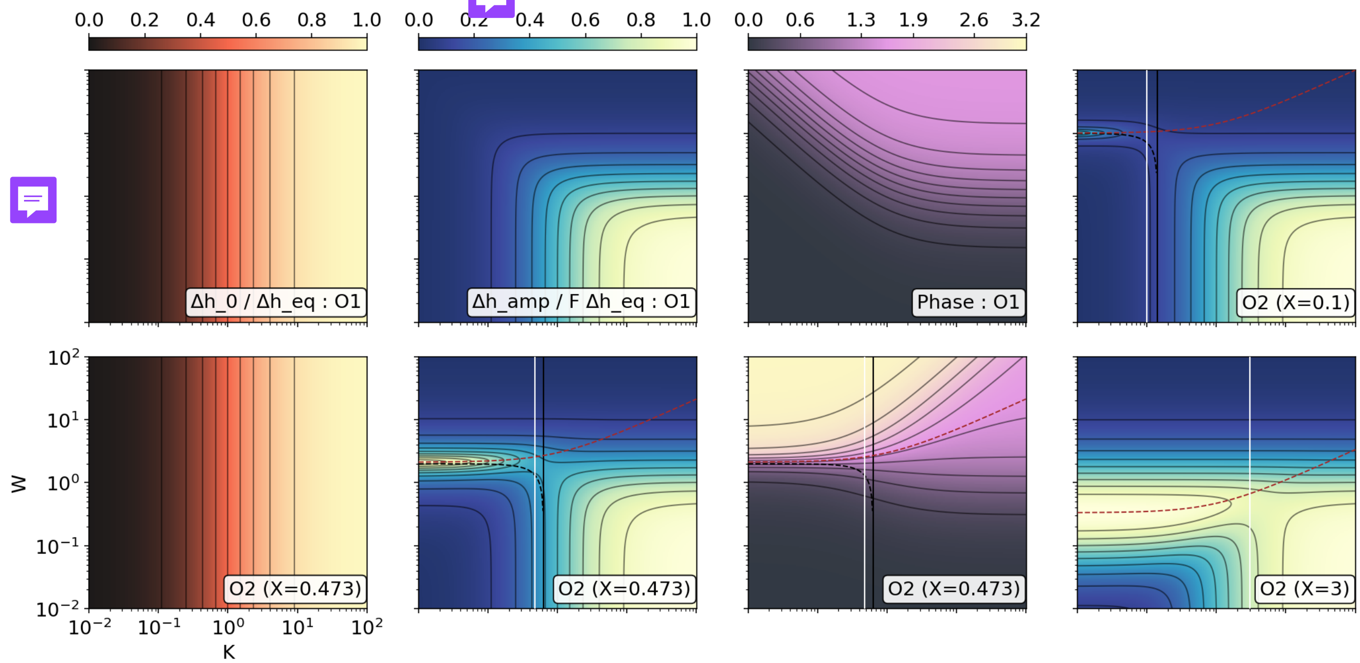

Figure 4. First and second order response of tidally-locked planets to variable heating, showing forcing normalized time-mean height magnitude ($\Delta h_{\mathrm{o}}/\Delta h_{\mathrm{eq}}$), its amplitude ($\Delta h_{\mathrm{amp}}/F\Delta h_{\mathrm{o}}$), and phase lag of the response using different colorschemes. The first order (O1) is dependent only on two nondimensional parameters $W = \omega\tau_{\mathrm{rad}}$ and K, while the second order (O2) has a third parameter $X = \tau_{\mathrm{wave}}/\tau_{\mathrm{rad}}$. The magnitude responses for both are the same and commensurate with D. Perez-Becker & A. P. Showman (2013). The brown and black dashed lines correspond to natural (Ω_{o}) and resonant (Ω_{res}) frequencies of the system. The solid black line corresponds to the critical damping K, to the left of which oscillatory solutions exist, and the system is overdamped to the right. The solid white line represents separation between the wave dominated inertial regime to the left of it and the drag/rotation dominated regime to the right, where the solutions match the first-order theory.

The nondimensional natural frequency of the system is $\Omega_{\mathrm{o}}\tau_{\mathrm{rad}} = \sqrt{(1+K)/X^2}$. Note that in this case, an additional independent dimensionless parameter (X) is required to define the space of the problem contrary to the two (K and W) in the first-order theory.

The panels labeled O2 in figure 4 show the time-mean height magnitude, its amplitude and, phase of planetary response for the second-order theory for different values of X. The magnitude of the response remains the same as captured by previous lower order theory and scaling relations, while the amplitude and phase show significant differences. The first thing to note is the region in $K - W$ space where the inertial physics (second-order theory) dominates. This is demarcated by the solid white line $K = X$, which is equivalent to the condition in equation (20) in dimensionless form. To the left of the white line, $K < X$ or $\tau_{\mathrm{wave}} < \tau_{\mathrm{eff}}$, i.e. gravity waves traverse the planet before they are damped. This supports the wave-driven resonance. Conversely, for $K > X$ (right of the white line) the wave timescale ~~to~~ large and the drag ~~too strong.~~ The white line thus separates a wave-active inertial regime (left) from a drag-dominated balanced regime (right) where the first-order theory is sufficient.

Second, the inertial regime ($K << X$) is seen to feature a local maxima in frequency space corresponding to ~~the~~ resonance. When K tends to 0, we ~~get~~ the following expression for the dimensionless natural frequency: to $W_{\mathrm{o}} = 1/X$. The correspondence can be seen at low values of K for respective X values in the amplitude plots in Figure 4 where the resonance originates. The brown dashed lines represent the natural frequencies of the system. Finally, the phase lag of the system can now take up values larger than $\pi/2$ as the theoretical limit reaches π symbolizing a complete out-of-phase response. This is typical to the behavior of second order systems with a restoring force. However, as we shall see later in simulations with appended discussion in §4.1.2, the phase lag never goes beyond $\pi/2$ due to the presence of nonlinearities in the system.

Note on the form of forcing? The forcing term in the second-order theory is of the form $\Delta h_{\mathrm{eq}}[\dot{f}/\tau_{\mathrm{rad}} + f/(\tau_{\mathrm{rad}}\tau_{\mathrm{eff}})]$ which is different from the first-order theory where it is simply $\Delta h_{\mathrm{eq}} f/\tau_{\mathrm{rad}}$. This is because the second-order theory retains the unsteady term in the momentum equation, which introduces an additional time derivative of the forcing function $f(t)$ in the governing equation. This leads to a more complex response of the system to time-variable irradiation, capturing both the direct effect of the forcing and its rate of change.

Thus, the response of tidally locked planets to time-variable irradiation can be described at two levels depending on whether inertia is dynamically active. Retaining local acceleration in the momentum equation leads to a formally second-order system in which the planetary heat contrast Δh behaves as a damped harmonic oscillator. In this regime, gravity-wave adjustment provides a restoring force while radiative relaxation and drag primarily provide damping. In the next section we explore through nonlinear simulations the role of nonlinearities in the system and how they deviate from the linearized theory presented above.

2.4. *Assumptions and limitations of the analytical framework*

The analytical results rest on several simplifying assumptions. Firstly, the effective drag approximation combines the Coriolis term and Rayleigh drag into a scalar $\tau_{\mathrm{eff}}^{-1} = f \sin\phi_0 + \tau_{\mathrm{drag}}^{-1}$ adequately estimates global heat redistribution (D. Perez-Becker & A. P. Showman 2013), but discards the directional information needed to predict the eastward vs. westward direction of hotspot oscillations. The simulations retain the full Coriolis term and reproduce this transition. Second, we assume a cosine ansatz for the trial solution $h'(x,t) = \Delta h(t)\cos(kx)$ that is symmetric about the substellar point implying purely substellar to antistellar point flow, so the theory does not predict any spatial hotspot displacement. The actual offset arises from the 2D Matsuno-Gill response (M. Hammond & R. T. Pierrehumbert (2018a), which is not included in this 1D framework. Third, we retain only the gravest wave mode $k = 1/a$ spanning the entire planet. At fast rotation, the Rossby deformation radius $L_d = c/\sqrt{f} \ll a$ and higher harmonics become energetically important, causing the theory to overestimate the planetary response (see simulation results later). Fourth, the meridional gradients are simplified by assuming periodic boundary conditions along longitudes, reducing the problem to a 1D zonal diffusion equation. This equatorial-strip approximation breaks down at fast rotation where off-equatorial Rossby wave structure becomes significant. Finally, the linearized theory neglects the term $\mathbf{R}$ (equation 1) which is responsible for equatorial superrotation (A. P. Showman & L. M. Polvani (2011), is omitted from linear calculations. The theory therefore cannot predict the equatorial jet, but the large-scale pressure-drag balance remains a reasonable approximation even in its presence.

Despite these limitations, the large-scale thermodynamic response — including the day–night contrast, oscillation amplitude, and phase lag — is well captured by the theory, as confirmed by the simulations in the following section.

3. NUMERICAL SIMULATIONS

To evaluate the efficacy of our linearized theory, we compare the analytical results presented before to linear and nonlinear hydrodynamic simulations using the full shallow water model.

3.1. *Framework*

Equations (1) and (2) are solved numerically using Dedalus3 (K. J. Burns et al. 2020), a flexible pseudo-spectral code for symbolic equation entry. We discretize the 2-D domain on a sphere basis and evolve the system using an SBDF2 timestepper. The simulation resolution is 32×16 (longitude $\times$ latitude), which, while low, sufficiently captures the core physical phenomena. This resolution corresponds to a minimum resolvable horizontal scale of $\sim$1000 km for an Earth-like planet — adequate for the dominant wavenumber-1 day–night response but insufficient to resolve structures smaller than the equatorial Rossby deformation radius at the fastest rotation rates explored. We admit that increasing the resolution would improve the accuracy of the simulations, but avoided doing so in the interest of exploring a larger parameter space. Our code was verified by reproducing results from existing literature (D. Perez-Becker & A. P. Showman 2013; J. Penn & G. K. Vallis 2017; K. Ohno & X. Zhang 2019) and is structurally the same as in D. Banik (2025).

The tidally locked height field is relaxed towards an equilibrium profile h_{eq} as shown in Figure 5, the intensity of which is periodically varied. The form of the equilibrium height field is given by

$$h_{\mathrm{eq}}(\lambda, \phi, t) = H + \Delta h_{\mathrm{eq}} \cdot \frac{\mathcal{R} + |\mathcal{R}|}{2}\,(1 + F\cos\omega t) \quad \text{where} \;\; \mathcal{R} = \cos\lambda\cos\phi. \tag{24}$$

Here, H is the nightside thickness, $\Delta \mathrm{h}_{\mathrm{eq}}$ is the forcing day-night contrast, and F and ω are the instellation fluctuation amplitude and frequency. The factor $(\mathcal{R} + |\mathcal{R}|)/2$ isolates hemispheric irradiation (A. R. Dobrovolskis 2009).

3.2. *Diagnostics*

Each run produces timeseries of global height and velocity fields. These are post processed to obtain the quantities that can be compared with the theory above. From the global height field we obtain the timeseries of the *maximum* height, the *mean* height and, the *hemispheric averaged* height. The time-mean magnitude of

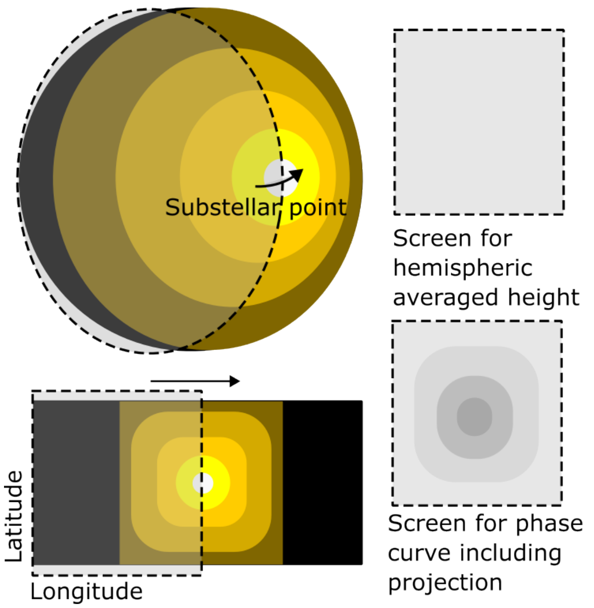

Figure 5. Typical equilibrium temperature distribution of a tidally-locked planet in two different representations, spherical and rectangular, showing the substellar point at the centre of the day side, and hemispheric screens for calculating the hemispheric averaged height and phase curves. The rectangular representation also shows the latitude, longitude and the night side. The arrows show the direction in which the screen moves to calculate the respective quantities. The screen for hemispheric averaged height is plain, while the second includes a projection to account for ray tracing effects needed for phase ~~calculations~~.

planetary response is calculated by taking the average of the peak and trough of the *maximum* height timeseries after the simulation is out of it's transient evolution phase and relaxed to a oscillating steady state. The amplitude is modulus half the difference between the peak and trough of the same. The statistical steady state is identified by tracking the time-averaged kinetic energy of the flow to have reached a constant value. Finally, the phase is obtained by calculating the temporal offset between the peaks in instellation and *mean* height and converting it to radians. The reason for choosing the mean height here is that it has a smoother sinusoidal structure than maximum height, especially for inertia dominated cases where interacting waves interfere to create complicated responses. The hemispheric averaged height is calculated as an average of the height field over a longitudinal extent of π the face of the planet centered at each individual longitude, and is a more accurate metric when comparing with observations that average the light coming from the exposed planetary hemisphere. We also obtain the longitudinal location of maximum height and the maximum hemispheric averaged height to track the motion of the real and observed hotspots.

3.3. *Exploration parameters*

We simulate 25 planets that span the entire nondimensional $K-W$ parameter space sampling 5 K values for each of 5 W values. We use two different magnitudes for the forcing $\Delta h_{\mathrm{eq}} = 10^{-3}$ and 1, respectively. F, the fractional amplitude is kept at 0.5. These parameter values cover the linear and nonlinear regime completely. We run these in two batches with $X = 0.528$ and 3, defining different ratios of radiative and wave adjustment strengths, as in figure 2. The first corresponds to smaller Earth-like planets, while the later uses the parameter set in D. Perez-Becker & A. P. Showman (2013) applicable to hot Jupiters.

Here, we recall that in the linearized theory we have assumed the Coriolis and drag terms to be combined into the effective drag term for simplification. However, their effects on the global flow patterns are distinct due to the nature of the individual terms — the Coriolis leads to latitudinal variations being stronger closer to the poles and weaker at the equator, while the drag depends only on the velocity ~~magnitude~~. For this reason, we vary K by varying both rotation and drag individually, to study if contrast properties are actually significantly affected by them. Finally, we vary W by changing the instellation variation frequency. All permutations included, we have a suite of 200 simulations ($25 \times 2 \times 2 \times 2$), the results of which are presented below.

3.4. *Typical Earth-like planets*

In this section we present results for the Earth-analog parameter set (Scenario A, Table 1; $g = 9.81\,\mathrm{m\,s^{-2}}$, $a = 6400\,\mathrm{km}$, $H = 100\,\mathrm{m}$, $\tau_{\mathrm{rad}} = 5\,\mathrm{days}$, $\tau_{\mathrm{wave}} = 2.365\,\mathrm{days}$, $X = 0.473$), which spans the linear regime for most of the K–W space explored, with the exception of the lowest K cases at $\Delta h_{\mathrm{eq}} = H$; see figure 2. Note: the theory plots in figure 4 and the exploration parameter discussion in §3.3 quote $X = 0.528$, which was used when generating those plots. The table-derived value is $X = 0.473$; the $\sim$12% difference does not qualitatively change any conclusions but should be reconciled in a future revision.

Figure 6 shows the simulated planetary responses for the above set. The general trends match the order-one theoretical results. However, the region of resonance occurring at low K produces a local minima at $W = 2$, which is only a feature of the second-order theory. This effect is a little more pronounced in the bottom two rows where the forcing is three orders of magnitude stronger than the top two rows. We also see the constant part of the response to have a manifestation of the resonance effect; this is not captured by any of the linearized theo-

Table 1. Physical parameters and timescales corresponding to the simulations presented. The table shows a total of 100 simulations, each of which are conducted at two different values of $\Delta h_{\rm eq} = 10^{-3}H$ and H, representing the linear and nonlinear regimes described above, resulting in a total of 200 simulations.

Parameter	**Symbol**	**Earth-like**	**Hot-Jupiter-like**
Surface gravity	g	$9.81\,\mathrm{m \cdot s^{-2}}$	$10\,\mathrm{m \cdot s^{-2}}$
Scale height	H	$100\,\mathrm{m}$	$4 \times 10^5\,\mathrm{m}$
Radiative timescale	$\tau_{\rm rad}$	5 days	0.1 days
Instellation variation	$\tau_{\rm instel}$	$3.14 - 314$ days ($\times 5$)	$0.06 - 6.28$ days ($\times 5$)
Wave timescale (derived)	$\tau_{\rm wave}$	2.365 days	0.4745 days
Wave-to-radiative timescale ratio (derived)	X	0.473	4.745
Scenario	**Rotation Period ($P_{\rm rot}$)**	**Drag Timescale ($\tau_{\rm drag}$)**	**No. of sims**
A: Earth-like			
- Batch 1	140 days	$0.11 - 22.373$ days ($\times 5$)	25
- Batch 2	$0.7 - 140$ days	22.373 days ($\times 5$)	25
B: Hot-Jupiter			
- Batch 3	282 days	$0.22 - 45$ days ($\times 5$)	25
- Batch 4	$1.42 - 282$ days	45 days ($\times 5$)	25

ries and possibly is due to higher order wave mean-flow interactions.

An interesting phenomena is observed for some of the simulations when we track the location of the hotspot. The hotspot oscillates as it moves away from the substellar point towards the antistellar point and back proceeding from the *eastward* direction, leading to thermal variability not only in time, but also in space. When it happens, we find the hotspot to be oscillating at the forcing frequency. The maximum amplitude of the hotspot oscillation is plotted in the last column of figure 6. The oscillation is seen to correspond to the region of the $K - W$ parameter space where the second-order theory predicts the resonant maximization of temporal variability. This is expected from the damped oscillator model of the planetary response, even though this phenomena is a spatially varying one.

The primary difference between the cases with changing drag and rotation is also apparent in these plots. Both set systems show spatial amplitude maximization during the resonance. However, it is only the cases with changing rotation period that also feature *westward* oscillations. This behaviour is novel and is a key component of atmospheric variability for such planets, and will be investigated in detail in a later section.

3.5. *Typical hot Jupiters*

In this section we present results for the hot Jupiter parameter set (Scenario B, Table 1; $g = 10\,\mathrm{m\,s^{-2}}$, $a = 82{,}000\,\mathrm{km}$, $H = 4 \times 10^5\,\mathrm{m}$, $\tau_{\rm rad} = 0.1$ days, $\tau_{\rm wave} = 0.4745$ days, $X = 4.745$), taken from D. Perez-Becker & A. P. Showman (2013). Compared to the Earth-analog, $X = 4.745$ implies a significantly higher $\tau_{\rm wave}/\tau_{\rm rad}$ ratio. The entire K–W space is in the overdamped second-order regime, but the pseudo-resonance amplitude peak described above still appears near $W \approx 0.25$. Only the momentum linearity condition is violated at $\Delta h_{\rm eq} = H$; the continuity equation remains linear throughout (see red line in figure 2). Note: the theory plots in figure 4 use $X = 3$, which was the value employed when generating those plots. The table-derived value is $X = 4.745$; this ∼37% difference shifts the resonant frequency and the inertial/overdamped boundary and should be reconciled by regenerating the theory plots with $X = 4.745$.

Figure 7 shows the planetary response to variable irradiation for the set of hot Jupiter planets described above. Comparing with the theoretical plots corresponding $X = 3$, we find a good match with both the first and second-order theories. The resonant frequency has now moved down to a smaller value of approximately $W = 0.25$. The top two rows for which $\Delta h_{\rm eq}$ is $10^{-3} \times H$, are in the linear regime, while the bottom two are not. The linear simulations match the theory better ~~for obvious reasons~~ — for the bottom rows, the $\Delta h_{\rm o}/\Delta h_{\rm eq}$ plots have more averaged out values over the $K - W$ parameter space, although the contour lines show that the fundamental structure is preserved. This is because, several of our assumptions breakdown (like what?) in the nonlinear regime, and additional physics (like what?) comes in. This is also seen in the amplitude plots with values averaged out for low W values.

We also observe that the phase offset goes beyond 90°, but never too far beyond to 180° as is determined

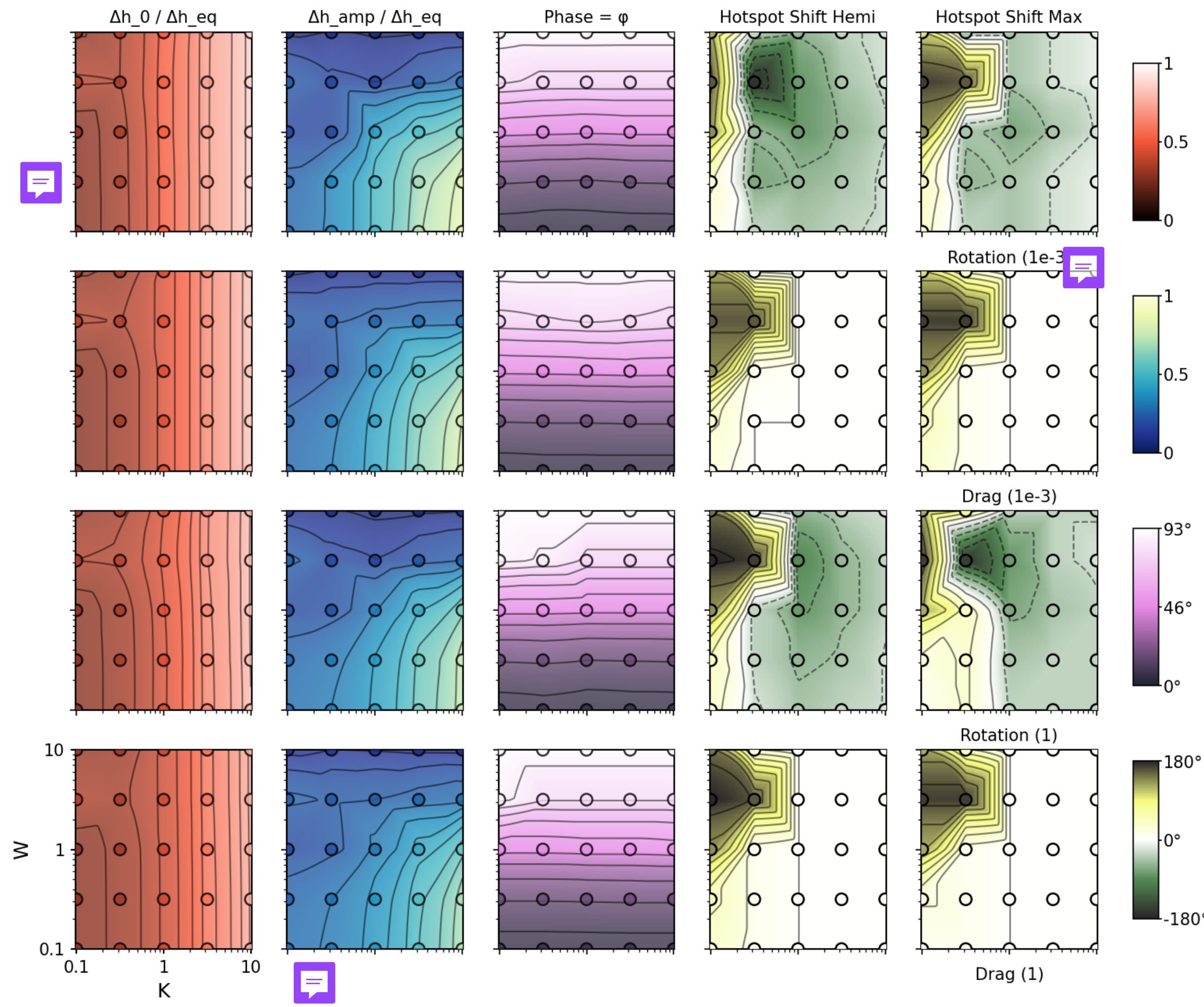

Figure 6. Suite of simulation results for an tidally-locked Earth-analog planet, showing the time-mean height magnitude, its amplitude, phase and maximum oscillating hotspot shift from the substellar point in the four columns, respectively. The colorschemes are identical to the ones used in theory plots in figure 4. The theoretical results capture the dominant features in the simulations including the second order resonance effects. Note that the magnitude plots feature a local maxima in frequency corresponding to the location of the resonance which is not seen in the theoretical results. Further, rotation and drag has overall similar effects for most quantities except for the hotsphot shift (4th column).

from the second order theory. (O. Shamir et al. 2023) shows that the steady, linearized Matsuno–Gill solution is only strictly valid when the temporal variation of the forcing is slow compared to the radiative/frictional relaxation time of the atmosphere – a requirement that traces back to the original steady-state assumption of T. Matsuno (1966) and A. E. Gill (1980). We notice an analogous effect in our simulations: at high instellation-oscillation frequency, the planetary response deviates from the purely sinusoidal form of the forcing towards more complicated forms (not shown here), even in nominally linear-amplitude runs. This likely leads to higher-order dynamics which prevents the phase lag of the planetary response at high instellation-variation frequency from reaching the theoretical maxima of 180° even in the otherwise-linear regime. The effects of nonlinearity by its intrinsic virtue is at multiple spatial and temporal scales, and hence the nonlinearity conditions in equations (10)–(9) may be regarded as sufficient, but not the only, conditions for nonlinear behaviour to emerge. Regardless, the large-scale flow and height patterns are still dominated by the linearized dynamics.

Finally, hotspot oscillations that appear in the resonant region are dominated by purely eastward move-

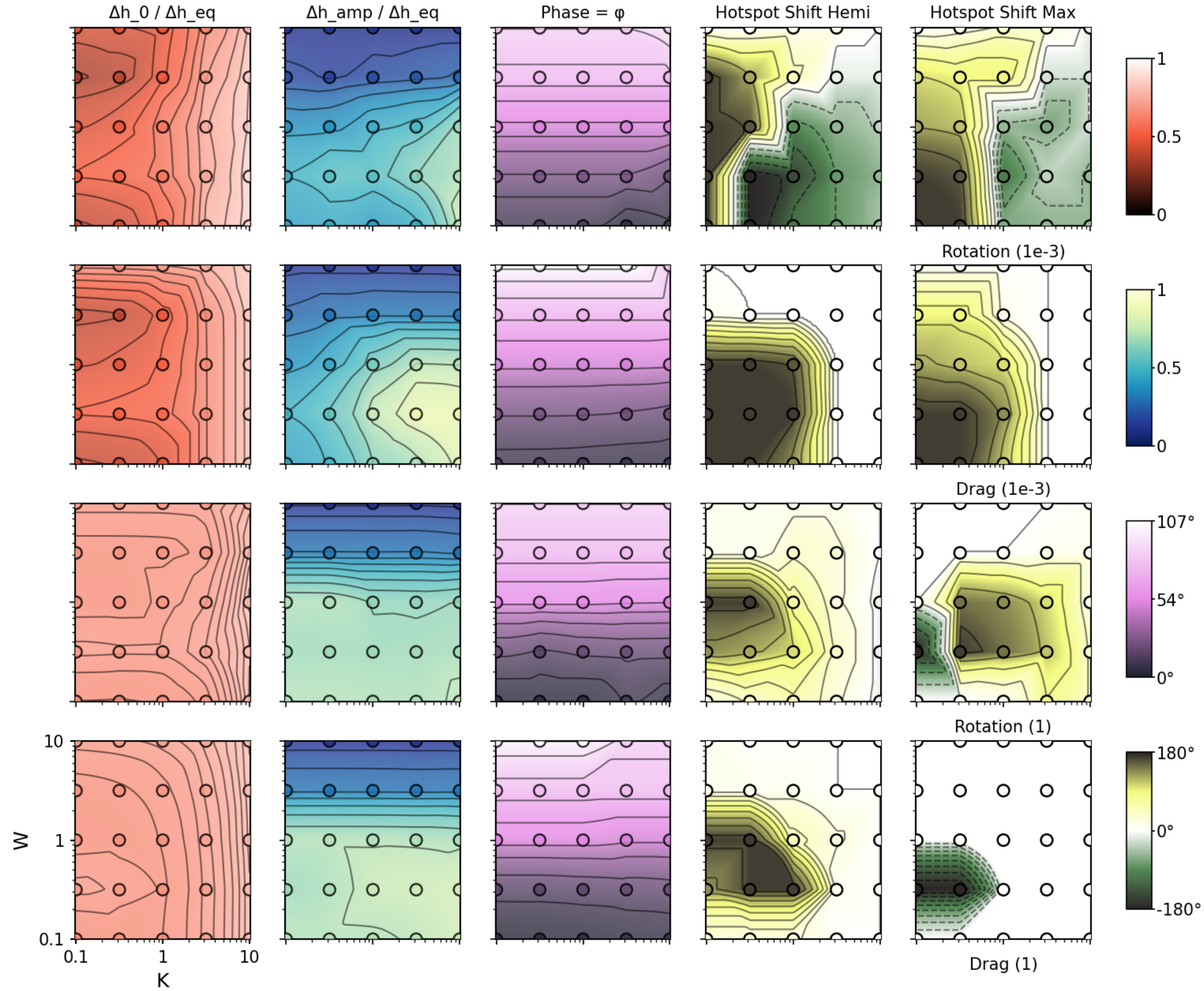

Figure 7. Suite of simulation results for an tidally-locked hot Jupiter planet, with the same quantities as in figure 6. Again, the theoretical results capture the dominant features in the simulations including the second order resonance effects. The magnitude plots still feature a local maxima in frequency corresponding to the location of the resonance which unseen in the theoretical results. Here, the differences between $\Delta h_{\rm eq} = 10^{-3} \times H$ and $1 \times H$ are stark and can be observed in most quantities presented. Notably different is the loss of westward (negative) hotspot oscillations in the strongly nonlinear regime with $\Delta h_{\rm eq} = 1H$ for cases with varying rotation (compare column 4, rows 1 and 3).This is likely because of the strengthening of the eastward equatorial jet caused by enhanced wave mean-flow interactions due to nonlinearities (cite papers here).

ment in the nonlinear regime. The change can be seen comparing column 4 panels in rows 1 and 3. Westward oscillations still appear in at low forcing strengths as it is in the linear regime. This is different from what we have seen before and explained in detail in a latter section.

Effect of F on plots must be discussed.

Overall, our simulations across Earth-analog and hot Jupiter parameter regimes demonstrate good agreement between theoretical predictions and SWM results for the first and second-order resonance features. Linear regime simulations match theory more closely, while nonlinear regimes exhibit phase offsets that show preferentially eastward-dominated hotspot motion. These results validate the theoretical framework; the rotation- and drag-dependence of hotspot oscillation direction is examined in detail in Section 4.

4. DISCUSSION

In this section we show the contrast between various forms of variability, specifically focussing on westward and eastward hotspot oscillations and probing the underlying mechanisms in the switch of this behaviour in linear and nonlinear regimes. We also produce synthetic phase curves for representative cases and show ~~an order-of-magnitude extent of variability in observable footprints of the hotspot oscillations.~~ We compare these with constant instellation equivalents.

Add literature for hotspot oscillations and contrast MHD with HD.

4.1. *Hotspot oscillations*

As seen previously in Figure 6 and 7, the direction and extent of hotspot oscillations depend on both the rotation rate and the drag timescale, even though the other theoretically derived response parameters such as magnitude, amplitude and phase-lag are agnostic to these variables. For slow rotating Earth-like and Hot-Jupiter like drag-varying batches, hotspot oscillations are purely eastward across all drag values. The extent of oscillation varies over the K–W parameter space with the maximum longitudinal shift (180 degrees) occuring at the location of the resonance and decaying in all other directions. Higher drag values lead to smaller oscillations, albeit justified by the dissipative nature of drag on wave or advection driven dynamics. The positive correlation between the maximum hotspot oscillation and the maximum planetary response amplitude at the resonance in the K–W parameter space shows that the phenomena are connected with ~~the~~ former ~~probably a~~ cause ~~of~~ the latter.

When rotation is varied, we find hotspot oscillations to be eastward at slow rotation and westward at fast rotation, with the maximum magnitude of oscillation still occuring at the resonance, for the Earth-like cases. Note here that this is primarily linear system. In most cases, the hotspot oscillates on only one side of the substellar point. However, in some, especially close to the transition rotation rate, the hotspot oscillates across the substellar point both westward and eastward. In Figures 6 and 7, such cases are still represented by the maximum hotspot oscillation thereby omitting this information, but the actual complex hotspot oscillation is better visualized in Hovmöller diagrams shown hereafter (Figure 8).

Finally, for the Hot-Jupiter cases in the nonlinear regime ($\Delta h_{\mathrm{eq}} = H$), we find that the oscillations are eastward even at fast rotation rates, making the behaviour consistent across the entire parameter space. This is a qualitative departure from linear behaviour and should have direct implications for observable phase curves for different planets as discussed in Section 4.2.

To visualise how rotation and nonlinearities alter the hotspot dynamics and track its movement, we construct Hovmöller diagrams — longitude–time maps of the height field — for the three representative linear and nonlinear Hot-Jupiter cases with changing rotation. Figure 8 compares linear ($\Delta h_{\mathrm{eq}} = 10^{-3}H$) and nonlinear ($\Delta h_{\mathrm{eq}} = H$) runs at three rotation periods (282, 4.9, and 1.42 days), showing the last three forcing cycles of each simulation. Two diagnostics are overlaid on every panel that show the meridional mean height averaged in *a window of 90 degrees width about the equator*, the hemispheric average height and their corresponding normalized values in order: the location of maximum height or the hotspot (white solid line) and the hemispheric-maximum hotspot (black dashed line). The dotted green line marks the substellar longitude. Note that the hemispheric average hotspot does not follow the same path as the meridional mean hotspot because they are different forms of averaging applied to the same height field. The latter is a better indicator of the astrometrically observable hotspot position owing to its projected surface average (see Figure 5).

The first thing to note is that the planetary response is strongly modulated by the instellation in all cases. This is due to the ~~low~~ radiative timescale of the hot Jupiter parameter set of 0.1 days. Second, the normalization carried out for each individual timestep and it highlights the instantaneous height maxima helping visualize the hotspot location. The absolute hotspot location matches well the maxima of the normalized meridional mean height, while the hemispheric-mximum hotspot matches the normalized hemispheric average height. Finally, the same conclusions are drawn from the Hovmöller diagrams as from the hotspot oscillation plots in Figures 6 and 7: linear cases show a transition from eastward to westward hotspot oscillations with increasing rotation with the extent of oscillation decreasing with rotation, while nonlinear cases remain eastward across all rotation rates.

4.1.1. *Role of rotation*

From the simulation results above, we note that the reflection of the resonance on the height response amplitude is rather minimal, but the hotspot oscillation is a more sensitive indicator. The resonance, if significant, always occurs at small values of rotation and drag (left hand side of the K–W plane). From linearized theory we know that strong rotation inhibits the transport of heat from the day to the night side via waves (see also simulations of (X. Tan & A. P. Showman 2020)), and so does strong drag. It is thus expected that the resonance and consequently hotspot oscillations will be the strongest when the wave propagation is least inhibited.

To understand the role of rotation let us consider slow rotation with *constant* instellation first. In this case, the

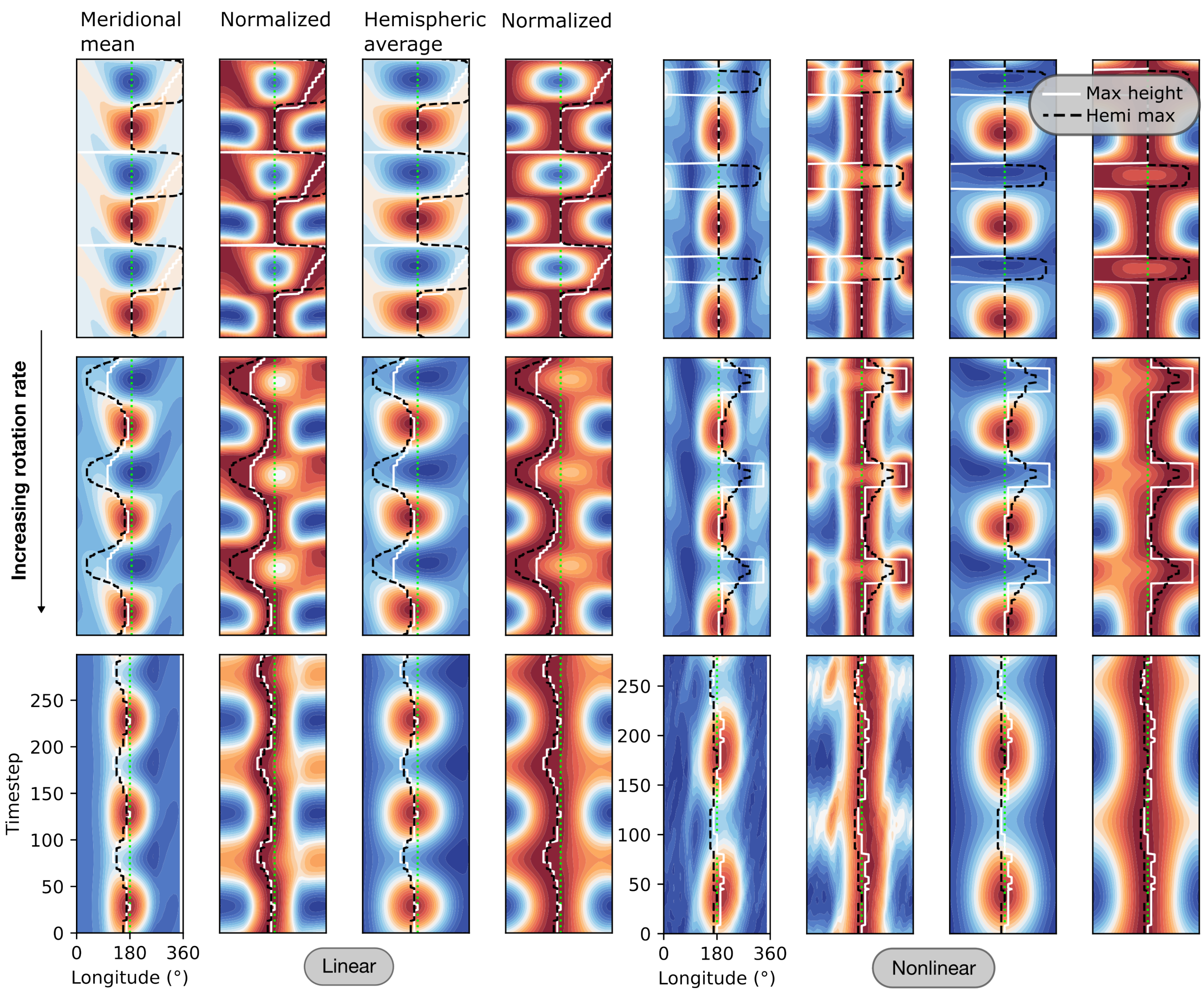

Figure 8. Hovmöller diagrams or timeseries of hotspot location relative to background zonal height profiles for six hot Jupiter planets showing hotspot oscillations switching from westward to eastward with rotation as the system transitions from linear to nonlinear regime. The left column corresponds to $\Delta h_{\rm eq} = 0.001$ and the ~~second~~ column to $\Delta h_{\rm eq} = 1$. Each row represents an increasing rotation period (282.9, 14.9 and 1.42 days), and columns show different diagnostic quantities: the meridional mean height, the merdional mean height normalized by the max height at each timestep to expose the instantaneous hotspot location, the hemispheric average height, and similarly normalized hemispheric average. The dotted green line marks the substellar longitude. The colorbars for background height fields are dropped to keep the figure clean focussing on the hotspot locations. White solid lines track the meridional mean-height hotspot, while black dashed lines track the hemispheric-maximum hotspot. The time window shown spans the last 3 forcing periods of the respective simulations.

day-night radiative forcing creates a standing equatorial Kelvin wave pattern, akin to the gravest gravity wave mode, that closely resembles the equilibrium height field but is not the same (A. E. Gill 1980; A. P. Showman & L. M. Polvani 2011). Since the Kelvin wave has an eastward propagating phase velocity, on varying the instellation, the system responds to the forcing by letting the Kelvin mode travel eastward till the instellation comes back to a maxima. Thus the point of maximum height, the hotspot, travels eastward and gets reset to the substellar point towards the end of the forcing cycle; see Figure 8, white line in the top left panels. The slowest rotation case is extremely close to the non-rotating case, where gravity wave propagation is isotropic from the substellar to the antistellar point. This is why we see the height fields to be almost symmetric about the substellar point even though the hotspot oscillation points to the east as a result of the eastward propagating weak Kelvin wave mode being excited by the instellation forcing.

As the rotation rate is increased (second and third rows of the left hand side panels of Figure 8 representing the linear regime), the Coriolis force becomes stronger. This leades to another off-equatorial wave mode, the Rossby wave, to be excited. The height fields are more asymmetric with the height maxima to the west (see second panel). The Rossby wave mode is westward propagating (T. Matsuno 1966; A. E. Gill 1980) and may ~~justify~~ the westward oscillation observed at fast rotation rates. However, in principle, both the Kelvin and Rossby waves are excited at all rotation rates according to linearized β-plane theory (T. Matsuno 1966; A. E. Gill 1980). Changing Ω on the β-plane only rescales the two wave solution in space and time — it does not change the relative amplitude of Kelvin vs. Rossby wave content, because that ratio is fixed by the forcing shape and damping alone in the scaled coordinates. Moreover, our system is on a sphere similar to the system in (O. Shamir et al. 2023). In their linearized case, even the fundamental wave modes like the eastward and westward inertio-gravity waves (EIG, WIG) and the mixed Rossby–gravity (MRG) wave are shown to be present. Each of these modes have their own direction of propagation and are excited at different degrees by the nature of forcing imposed, which is different between their study and ours. Additionally, even though we find that height contrast is agnostic to rotation and drag, O. Shamir et al. (2023) shows the wave mode decomposition is differently sensitive to these quantities and hence affects the hotspot location. Reducing rotation rate progressively breaks the two-wave (Kelvin+Rossby) picture altogether, ~~pulling weight into~~ inertio-gravity modes that ~~don't exist as a distinct, relevant category~~ in the pure equatorial β-plane theory. The exact mechanism for the switch in direction of hotspot oscillation is thus not clear and will be investigated in a future study.

4.1.2. *Role of nonlinearities*

We now study the role of nonlinearities, the conditions for which are given in equations (10) and (9). Simulations representing the nonlinear Hot-Jupiter regime ($\Delta h_{\mathrm{eq}} = H$) are based on these conditions. However, the conditions themselves are agnostic to the forcing frequency. We have noted that a high forcing frequency can play a role in the local rise of nonlinearities, but we do not resolve small scales here and hence the nonlinear behaviour is restricted to its effect on the global scale features. We emphasize that nonlinear behaviour is not synonymous with wave-driven behaviour, even though both formally arise from the same inertial terms in the momentum equations.

In the nonlinear regime at low rotation (1st row, right-hand-side panels of Figure 8), we see the onset of a switching behaviour of the hotspot between the substellar and antistellar points, in contrast to the gradual eastward drift seen in the corresponding linear simulations. This hints at the presence of higher-order interactions that are absent in the linear regime (A. P. Showman & L. M. Polvani 2010b, 2011). The exact mechanism responsible is not yet clear and warrants further investigation. However, the hotspot still oscillates eastward in this regime, similar to the linear case.

At intermediate rotation (2nd row, right-hand-side panels), the hotspot oscillations continue to occur in the eastward direction, in contrast to the westward oscillation seen in the corresponding linear case. We suggest that the suppression of westward oscillations in the nonlinear regime is a consequence of the wave–mean-flow mechanism of A. P. Showman & L. M. Polvani (2010b, 2011). In that mechanism, the equatorward eddy momentum flux convergence associated with the standing Kelvin–Rossby wave pattern spins up a persistent eastward equatorial superrotating jet. Once established, this jet Doppler-shifts the stationary forced wave pattern to produce an eastward hotspot offset in the absence of instellation variation (M. Hammond & R. T. Pierrehumbert 2018a). In the linear regime, by contrast, the wave-driven momentum flux convergence scales weakens and is unable to spin up or maintain an equatorial jet against drag. Without a jet to Doppler-shift the wave pattern, the Rossby-eddy pattern is free to dominate at fast rotation, producing the westward oscillations described above. We note, following Q. Nicolas & G. K. Vallis (2026), that the strength of superrotation and the magnitude/direction of the resulting hotspot offset are not related in a simple, one-to-one way, but are jointly controlled by rotation, drag, and radiative timescale; the argument above should therefore be regarded as a plausible leading-order mechanism rather than a complete explanation.

Finally, at high rotation rates (3rd row, right-hand-side panels), the hotspot offset – while still eastward – is reduced, as rotation inhibits the day-to-nightside transfer of heat, similar to the 3D GCM simulations of X. Tan & A. P. Showman (2020), who also find the eastward offset to shrink as rotation increases.

4.1.3. *Caveats and further exploration*

The interpretation above rests on a qualitative argument for how wave–mean-flow interactions might suppress the westward, Rossby-dominated pattern found in the linear regime, but we have not yet directly diagnosed the jet or decomposed the flow into its constituent wave

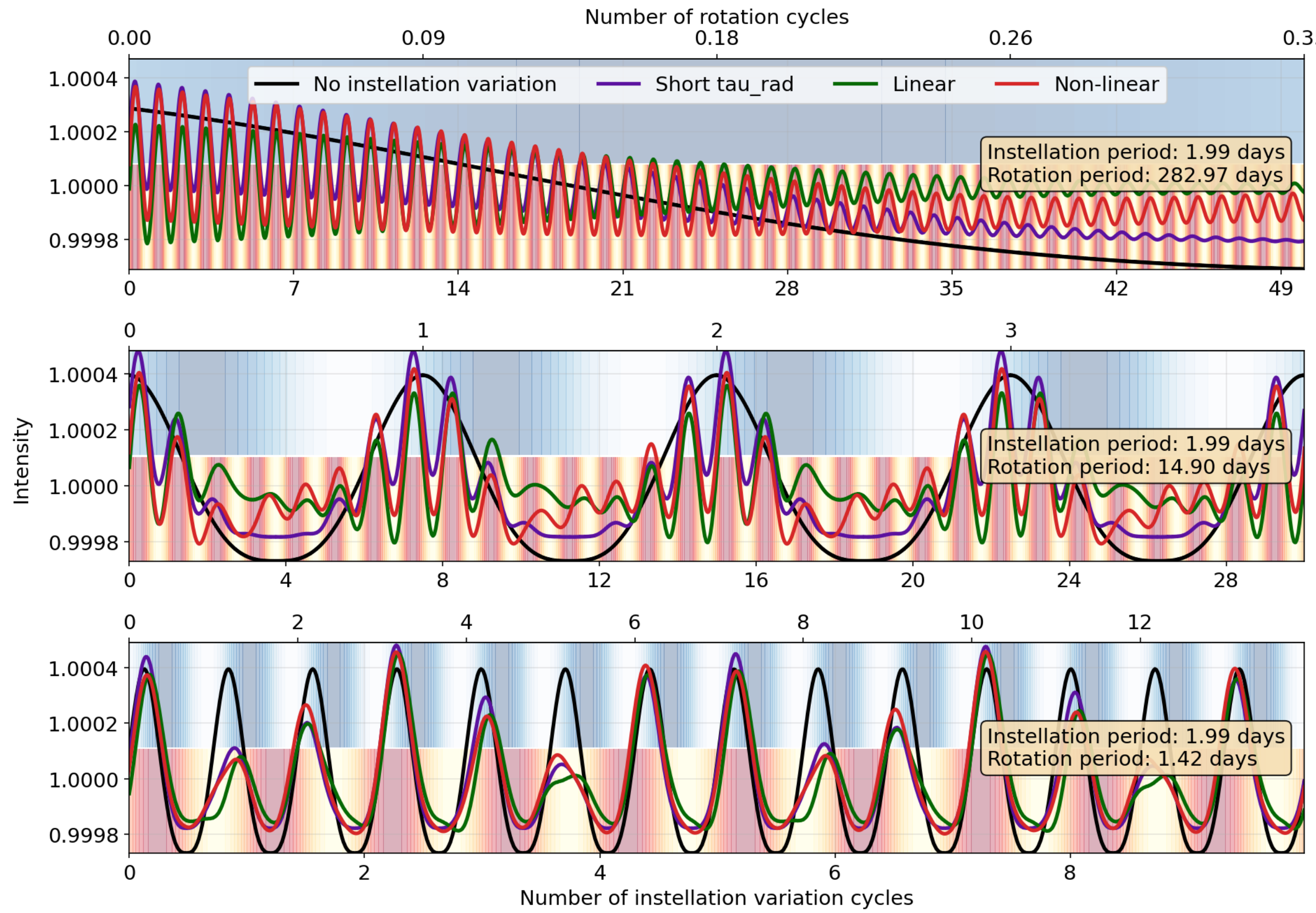

Figure 9. Phase curves for the same rotation periods as in Figure 8 with linear and nonlinear cases show significant differences. Each panel shows four normalized phase curves: case for no instellation variation, or simply the phase curve of the forcing profile (black); the same with instellation variation (purple); the linear case (green); and the nonlinear case (red). The background contours represent instellation variations (yellow-red, lower half) and rotation variations (blue and white, upper half) with corresponding cycle number on the top and bottom axes.

modes. Two directions of investigation would sharpen this picture considerably. First, an explicit projection of both the linear and nonlinear solutions onto the free wave modes of the system, following the methodology of O. Shamir et al. (2023), would let us test directly the mechanism of hotspot behaviour in the linear regime. Second, in the nonlinear regime, we may probe through a time-dependent helmholtz decomposition of the flow field (M. Hammond & N. T. Lewis 2021), the contributions of the divergent, eddy-rotational and eddy-mean components and the timescales over which they grow and decay subject to transient instellation to better understand how the equatorial jet spins up and equilibrates (A. P. Showman & L. M. Polvani 2011) thereby delineating the hotspot behaviour. Disentangling these transient jet dynamics from the quasi-steady wave response is left for future work.

4.2. *Synthetic phase curves and implications for observations*

The disc-integrated flux — the observable phase curve — is constructed by integrating the hemispheric-averaged height field over the visible hemisphere as a function of orbital phase. Figure 9 shows phase curves for the same six rotation periods as the Hovmöller analysis. Each panel overlays four normalised curves: a baseline with no instellation variation (black); a constant-instellation run with orbital phase sampling (purple); the linear variability case (green); and the nonlinear variability case (red). Background shading encodes the instellation cycle number, allowing direct comparison of the forcing periodicity and the atmospheric response.

The first thing to note in Figure 9 is that both the rotation and instellation variation modes crop up distinctly in the phase curves when the periods are significantly separated from each other as in the first two

Table 2. Representative systems with variable irradiation placed in K–W parameter space. $\tau_{\rm rad}$ is estimated via $\tau_{\rm rad} \approx 0.1\,(1600\ {\rm K}/T_{\rm eq})^3$ days (D. Perez-Becker & A. P. Showman 2013); K carries an additional order-of-magnitude uncertainty from the drag timescale. Categories: A = eccentric orbit; B = gravity-darkened host; C = pulsating host; D = circumbinary orbit.

System	Cat.	$T_{\rm eq}$ (K)	$\tau_{\rm rad}$ (d)	$T_{\rm force}$ (d)	W	K	F	Regime
WASP-33b	C	3200	0.013	0.008–0.08	1–6	$\sim$1	$\ll 0.01$	near-resonant
KELT-9b	B	4050	0.006	0.74	0.04	$\sim$1	$\sim$0.1	quasi-steady
KELT-20b	B	2345	0.022	$\sim$0.45	0.31	$\sim$1	$\sim$0.05	transitional
HD 80606b	A	1200	0.19	111.4	0.011	$\sim$0.3	$\sim$1	quasi-steady, nonlinear
HAT-P-2b	A	1449	0.12	5.63	0.13	$\sim$0.5	0.28	transitional
HD 17156b	A	1322	0.18	21.2	0.053	$\sim$0.4	0.40	quasi-steady
HAT-P-34b	A	1611	0.10	5.26	0.12	$\sim$0.5	0.29	quasi-steady
Kepler-47b	D	$\sim$200	$\gg 1$	7.4	$\gg 1$	$\ll 1$	0.05–0.10	muted
TOI-1338b	D	$\sim$260	$\gg 1$	14.6	$\gg 1$	$\ll 1$	$\sim$0.10	muted
Kepler-35b	D	$\sim$202	$\gg 1$	20.7	$\gg 1$	$\ll 1$	$\sim$0.10	muted

panels, as opposed to when they are similar as in the last panel. In this case, we find the overall variation to match the rotation better than the instellation variation, although the relative secondary eclipse heights ~~is~~ strongly modulated by the instantaneously available instellation; smaller peaks in yellow regions (low instellation) and larger peaks in red (strong instellation). Also, the eclipsing phase (lowest points) of the instellation variation cases never hits the lowest of the constant instellation case because the hotspot oscillations averaged over time help increase the nightside temperature. The highest point is consequently higher, although not observed as prominently for the fastest rotating planet. The reason the fastest planet shows a different behavior is probably because it is out of the resonance regime where the planet response is well characterized by the first-order linearly damped model in §2.2.

The additional modulation of the phase curve introduced by instellation variability scales linearly with F, the fractional semi-amplitude. In the current work this value has been kept to 50% of the constant magnitude, which is not realistic for most systems. However, the qualitative behaviour of the phase curves is expected to remain similar.

Add JWST baseline comparison estimates.

4.3. *Where do planets fall in K-W space?*

The K–W diagram is not merely a theoretical tool; it provides a map on which real observed systems can be placed to predict the character of their atmospheric response to variable irradiation. Table 2 compiles ten representative systems spanning all four variability categories (A–D), together with order-of-magnitude estimates of the key timescales and nondimensional parameters. Radiative timescales are estimated via $\tau_{\rm rad} \approx 0.1\,(1600\ {\rm K}/T_{\rm eq})^3$ days (D. Perez-Becker & A. P. Showman 2013); values of K carry an additional order-of-magnitude uncertainty from the unknown drag timescale $\tau_{\rm drag}$. The systems span more than four orders of magnitude in W and occupy all dynamical regimes: WASP-33b ($W \sim 1$–6) lies near or above resonance; KELT-20b and HAT-P-2b are in the transitional range ($W \sim 0.1$–0.3); KELT-9b, HD 17156b, HAT-P-34b, and HD 80606b fall in the quasi-steady limit ($W \ll 1$); and the cold circumbinary planets (Kepler-47b, TOI-1338b, Kepler-35b) all have $W \gg 1$ and are expected to show muted atmospheric responses to binary forcing. These placements are illustrated in Figure 10.

Note that $T_{\rm force}$ for WASP-33b is the stellar pulsation period; for KELT-9b it is $T_{\rm orb}/2$ (both stellar poles encountered per orbit); for KELT-20b it is the stellar rotation period; for eccentric planets it is the orbital period; for circumbinary planets it is the inner binary eclipse period.

5. CONCLUSION

We have presented an analytical and numerical study of the atmospheric response of tidally locked exoplanets to ~~temporally modulated~~ stellar irradiation~~,~~ using a forced shallow water model ~~as the dynamical framework~~. The central achievement of ~~the~~ linear theory is a dimensional reduction: the first-order (drag-dominated) response is fully characterised by two nondimensional parameters, $K = \tau_{\rm wave}^2/(\tau_{\rm rad}\tau_{\rm eff})$ and $W = \omega\tau_{\rm rad}$ —

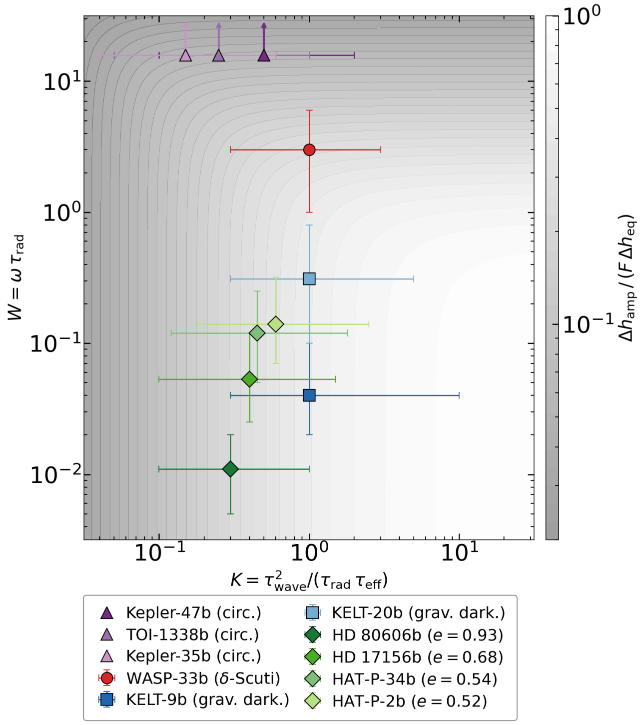

Figure 10. Placement of ten variably irradiated systems (Table 2) in the K–W parameter space, overlaid on the first-order amplitude response $\Delta h_{\rm amp}/(F\, \Delta h_{\rm eq})$ (colour scale). Marker shapes encode variability category: circles (C, pulsating), squares (B, gravity-darkened), diamonds (A, eccentric), upward triangles (D, circumbinary). Error bars reflect the order-of-magnitude uncertainty in K from the unknown drag timescale. The three circumbinary planets (Kepler-47b, TOI-1338b, Kepler-35b) all have $W \gg 1$ and are shown as upward arrows at the chart boundary. WASP-33b ($W \sim 1$–6) lies near or above resonance; KELT-20b and HAT-P-2b are in the transitional regime; the remaining hot Jupiters lie in the quasi-steady ($W \ll 1$) corner.

encoding, respectively, the day side heat retention efficiency and the intellation oscillation — from which closed-form expressions for the mean height contrast, oscillation amplitude, and phase lag follow directly.

In the steady-forcing limit ($W \to 0$), the first-order amplitude expression reduces to $\Delta h/\Delta h_{\rm eq} = (1 + 1/K)^{-1}$, recovering and generalising the heat redistribution scaling of D. Perez-Becker & A. P. Showman (2013) and T. D. Komacek & A. P. Showman (2016) to the variable-irradiation ~~setting~~. For high heat redistribution ($K \ll 1$), inertial/wave ~~physics dominate~~ and a second-order resonance amplifies the atmospheric response. The dimensionless natural frequency is lower for hot Jupiters ($X \gg 1$) than for Earth-like planets ($X < 1$), reflecting the faster wave adjustment of the former relative to their radiative timescale, and is reproduced qualitatively in both linear and nonlinear simulations. This resonance has been identified for the first time in studies of tidally locked planets (J. Penn & G. K. Vallis 2017; M. Hammond & R. T. Pierrehumbert 2018b), and constitutes the primary new theoretical result of this work.

A second key result concerns the direction of hotspot oscillations, which we find to be controlled primarily by rotation rate rather than drag timescale: drag-varying experiments at fixed (slow) rotation show purely eastward oscillation, while increasing rotation in rotation-varying experiments drives a transition from eastward to westward oscillations. This transition is suppressed in the nonlinear regime ($\Delta h_{\rm eq} \sim H$), where nonlinear momentum advection drives all oscillations eastward regardless of rotation rate — consistent with the wave–mean-flow and jet spin-up mechanism of M. Hammond & R. T. Pierrehumbert (2018b) and S.-M. Tsai et al. (2014), in which the equatorial superrotation causes the hotspot to shift eastwards (A. P. Showman & L. M. Polvani 2011).

A paragraph on the observational implications of the results should be added here.

Several avenues remain for future work. The shallow water model neglects vertical structure, baroclinic instability, magnetic drag, and cloud feedbacks, all of which can modify phase curve morphology (R. Perna et al. 2010; M. Wazny & K. Menou 2024); 3D GCM simulations will be needed to test whether the resonance and oscillation-direction predictions survive in more realistic atmospheres. The mechanism underlying the westward oscillations in the linear, fast-rotation regime and their nonlinear suppression warrants further study in terms of angular momentum budget and wave–mean flow interaction. Finally, mapping observed systems (WASP-33b, KELT-9b, HD 20782, Kepler-47b) onto the K–W plane will identify the best candidates for detecting variable-irradiation signatures with JWST.

Software: Claude, Gemini, ChatGPT, NumPy, SciPy

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